

Technical Datasheet — VBF-T6R1-DSH-01

LNMO/graphite high-voltage smartphone cell — virtual design (DFN simulation fidelity)

Field	Value	Source
Rated capacity	6.20 Ah (1C, simulation-verified); parameter-set nominal 4.5 Ah	cell/r8_d6_1c_dfn.json:capacity_ah; parameter set
Nominal voltage / window	4.11 V plateau midpoint; window 2.5 – 4.7 V	cell/r8_d6_energy_dfn.json:midpoint_voltage_v; parameter set
Rated energy	25.41 Wh	cell/r8_d6_energy_dfn.json:energy_wh
Volumetric energy density	1135.8 Wh/L	cell/r8_d6_energy_dfn.json:energy_density_wh_l
Gravimetric energy density	493.3 Wh/kg (electrolyte excluded — parameter set lacks density)	cell/r8_d6_energy_dfn.json:energy_density_wh_kg
Maximum continuous discharge rate	1C verified (6.198 Ah); 5C / low-temperature retention: not provided (outside contract; protocol exists, not run)	cell/r8_d6_1c_dfn.json
Fast-charge capability	4C CC-CV charge at 45 °C ambient: T_max = 321.42 K (48.27 °C) ≤ 50 °C; no lithium plating (min anode potential +10.96 mV)	cell/r8_d6_4c_dfn.json
DC internal resistance	6.38 mΩ (calc-energy midpoint DCR)	cell/r8_d6_energy_dfn.json:dcr_ohm
Power density	13.52 kW/kg	cell/r8_d6_energy_dfn.json:power_density_w_kg
Operating temperature range	verified at 25 °C (1C discharge/aging) and 45 °C (4C charge); wider range not simulated	protocol definitions
Cycle life	100 × 1C cycles simulated: SEI thickness 385.3 nm ≤ 500 nm target ✓. Cycle count to end-of-life: Not simulated (no capacity-fade EOL criterion in contract)	cell/r8_d6_aging_dfn.json:sei_thickness_nm_end
Safety determination	No plating at 4C (margin 11.0 mV); T_max margin 1.73 K below 50 °C red line; cooling requirement h = 400 W/m²K (active cooling design)	cell/r8_d6_4c_dfn.json; params_final.json
Dimensions	65 mm × 1580 mm × 217.8 μm electrode stack; pouch shell thickness Not provided (not modeled)	parameter set; cell/r8_d6_energy_dfn.json
Mass	51.50 g (electrolyte excluded); ≈ 57.8 g with electrolyte design estimate	cell/r8_d6_energy_dfn.json:mass_kg; design_spec §3

Formulation notes (honesty): electrolyte diffusivity 6.0e-10 m²/s is a constant override at the upper bound of reported low-viscosity fast-charge carbonate formulations (Nyman2008 baseline ≈ 2.9e-10 at 1000 mol/m³); materializing the design requires a real low-viscosity electrolyte with measured D_e in this range. EC-lean electrolyte (EC 2270.5 mol/m³) plus EC diffusivity 1.0e-18 m²/s are the SEI-control levers — an additive/solvent package must reproduce these effective values. All values are tool-output; this datasheet is a virtual-design document, not a production datasheet.